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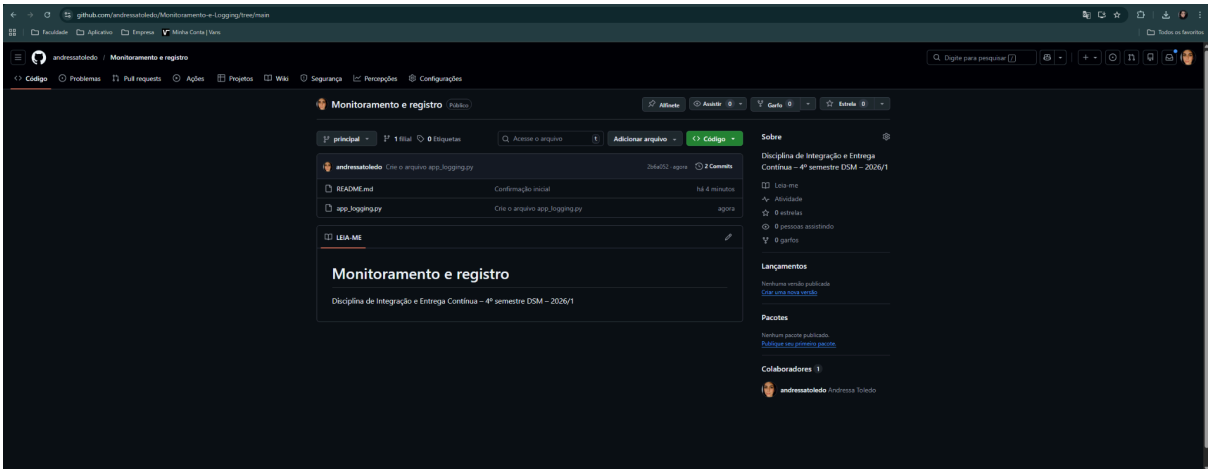
Matéria: Integração e Entrega Contínua - 2026/1

Atividade - Aula 06

Exercício 1



Exercício 2

```python
logger = logging.getLogger("app_logger")
handler = logging.StreamHandler()
handler.setFormatter(JSONFormatter())
logger.addHandler(handler)
logger.setLevel(logging.INFO)

REQUEST_TIME = Summary('request_processing_seconds', 'Tempo de processamento de requisição')

app = Flask(__name__)

@app.route('/')
@REQUEST_TIME.time()
def index():
    logger.info("Processando requisição na rota principal")
    time.sleep(random.uniform(0.1, 0.5))
    return "Aplicação Online!"

@app.route('/error')
def error():
    logger.error("Erro crítico detectado na rota de teste!")
    return "Erro simulado", 500

if __name__ == '__main__':
    start_http_server(8000)
    logger.info("Iniciando aplicação na porta 5000 e métricas na 8000...")
    app.run(port=5000)
```

```
PS C:\Users\Usuario\Desktop\Andressa Toledo\Faculdade\4 SEMESTRE\Integração e Entrega Contínua\Monitoramento-e-Logging> python app.py
{"timestamp": "2026-03-23 20:58:57,206", "level": "INFO", "message": "Iniciando aplica\u00e7\u00e3o na porta 5000 e m\u00e9tricas na 8000...", "module": "app", "service": "API-Pagamentos"}
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
{"timestamp": "2026-03-23 20:59:05,614", "level": "INFO", "message": "Processando requisi\u00e7\u00e3o na rota principal", "module": "app", "service": "API-Pagamentos"}
127.0.0.1 - - [23/Mar/2026 20:59:05] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [23/Mar/2026 20:59:05] "GET /favicon.ico HTTP/1.1" 404 -
{"timestamp": "2026-03-23 20:59:23,291", "level": "ERROR", "message": "Erro cr\u00edtico detectado na rota de teste!", "module": "app", "service": "API-Pagamentos"}
127.0.0.1 - - [23/Mar/2026 20:59:23] "GET /error HTTP/1.1" 500 -
127.0.0.1 - - [23/Mar/2026 20:59:23] "GET /favicon.ico HTTP/1.1" 404 -
{"timestamp": "2026-03-23 20:59:53,753", "level": "ERROR", "message": "Erro cr\u00edtico detectado na rota de teste!", "module": "app", "service": "API-Pagamentos"}
127.0.0.1 - - [23/Mar/2026 20:59:53] "GET /error HTTP/1.1" 500 -
```

Exercício 03

prometheus.yml U | logstash.conf U | app.py U

logstash.conf

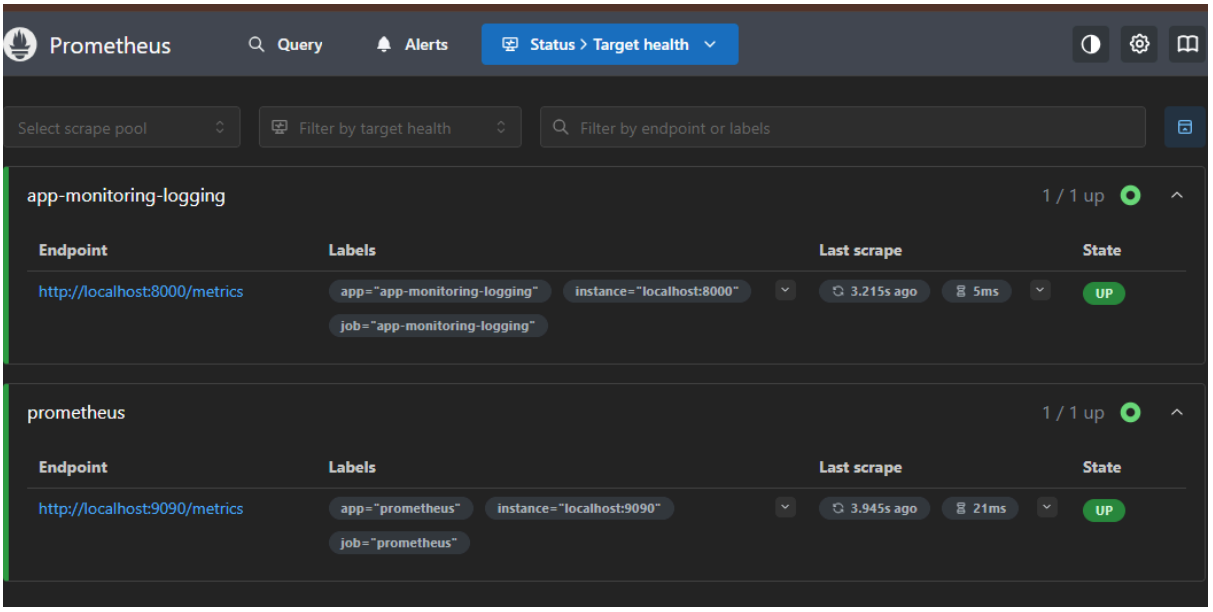
```
input {
  file {
    path => "/caminho/para/seu/projeto/app.log"
    start_position => "beginning"
  }
}

filter {
  json {
    source => "message"
  }
}

output {
  elasticsearch {
    hosts => ["http://localhost:9200"]
  }
  stdout { codec => rubydebug }
}
```

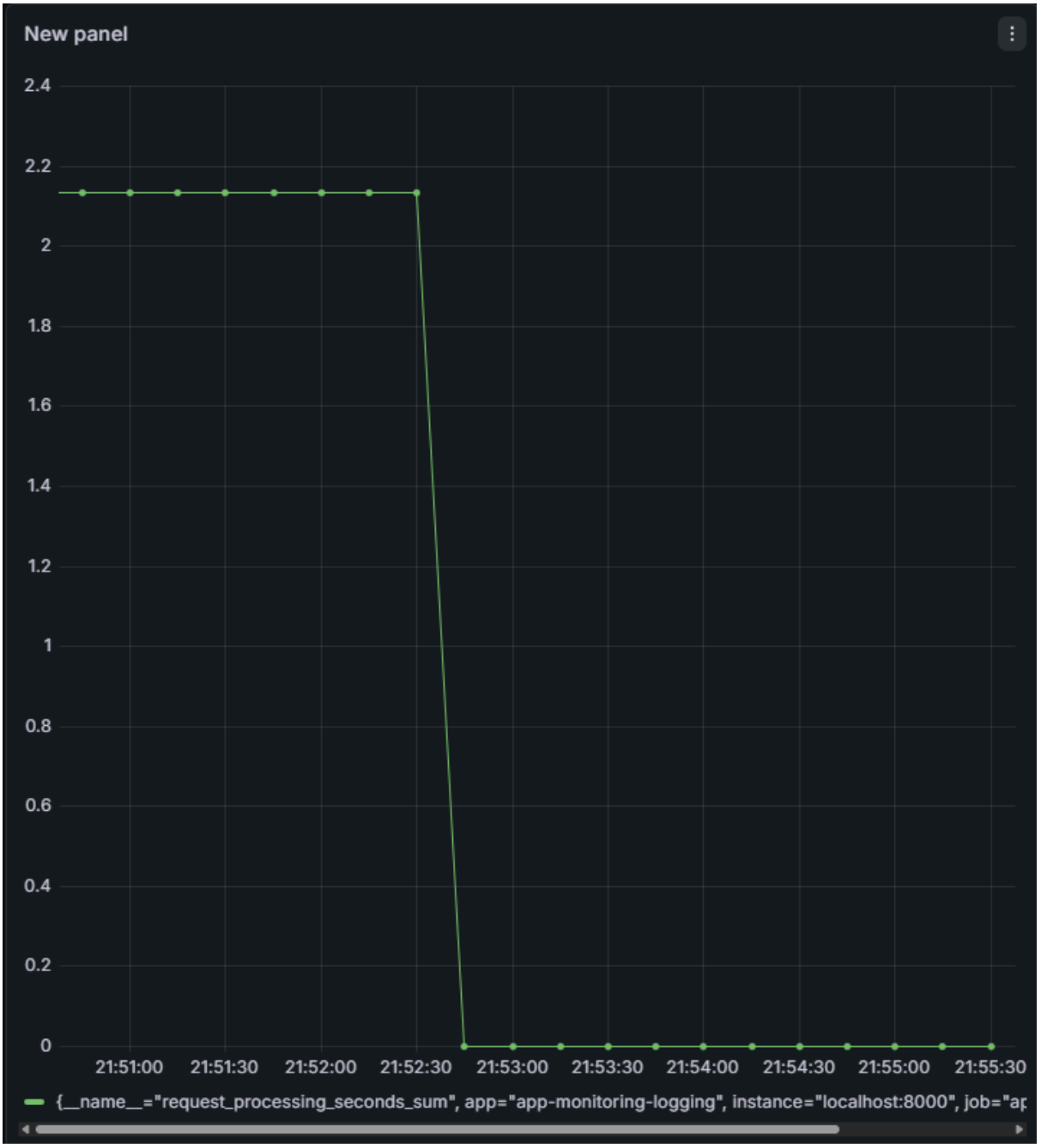
Exercício 4

localhost:8000/metrics

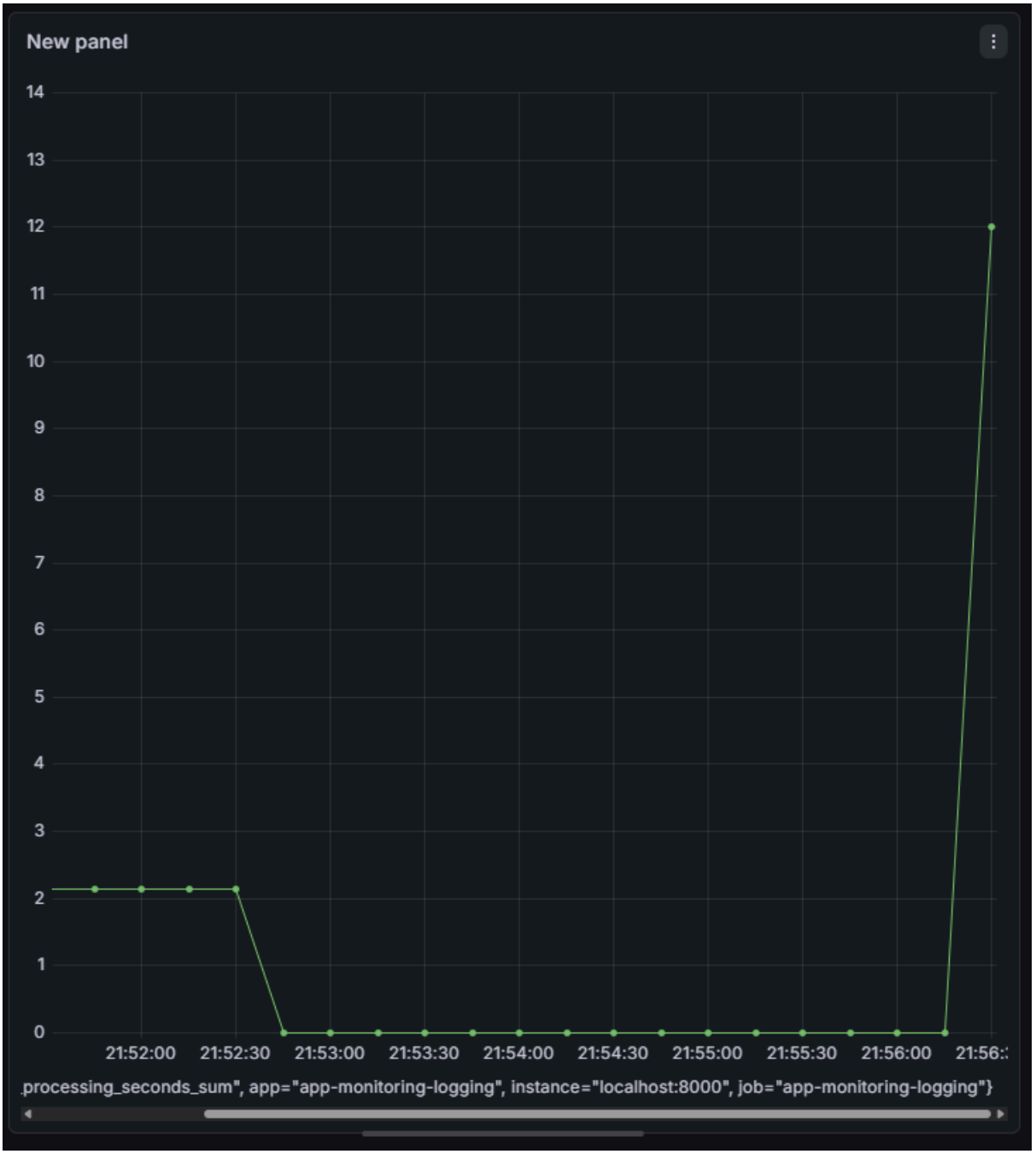
Caminhão - Google... | Inclusão de motoris...

```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 480.0
python_gc_objects_collected_total{generation="1"} 135.0
python_gc_objects_collected_total{generation="2"} 0.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 79.0
python_gc_collections_total{generation="1"} 7.0
python_gc_collections_total{generation="2"} 0.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="11",patchlevel="9",version="3.11.9"} 1.0
# HELP request_processing_seconds Tempo de processamento de requisição
# TYPE request_processing_seconds summary
request_processing_seconds_count 8.0
request_processing_seconds_sum 14.14082960000178
# HELP request_processing_seconds_created Tempo de processamento de requisição
# TYPE request_processing_seconds_created gauge
request_processing_seconds_created 1.774313539049282 3e+09
```


Prometheus
Query
Alerts
Status > Target health
Select scrape pool
Filter by target health
Filter by endpoint or labels
app-monitoring-logging
1 / 1 up
Endpoint
Labels
Last scrape
State
http://localhost:8000/metrics
app="app-monitoring-logging"
instance="localhost:8000"
job="app-monitoring-logging"
3.215s ago
5ms
UP
prometheus
1 / 1 up
Endpoint
Labels
Last scrape
State
http://localhost:9090/metrics
app="prometheus"
instance="localhost:9090"
job="prometheus"
3.945s ago
21ms
UP



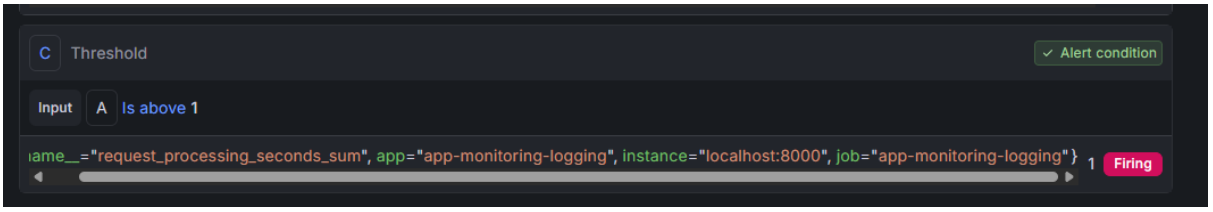
Funcionando com tempo de resposta normal

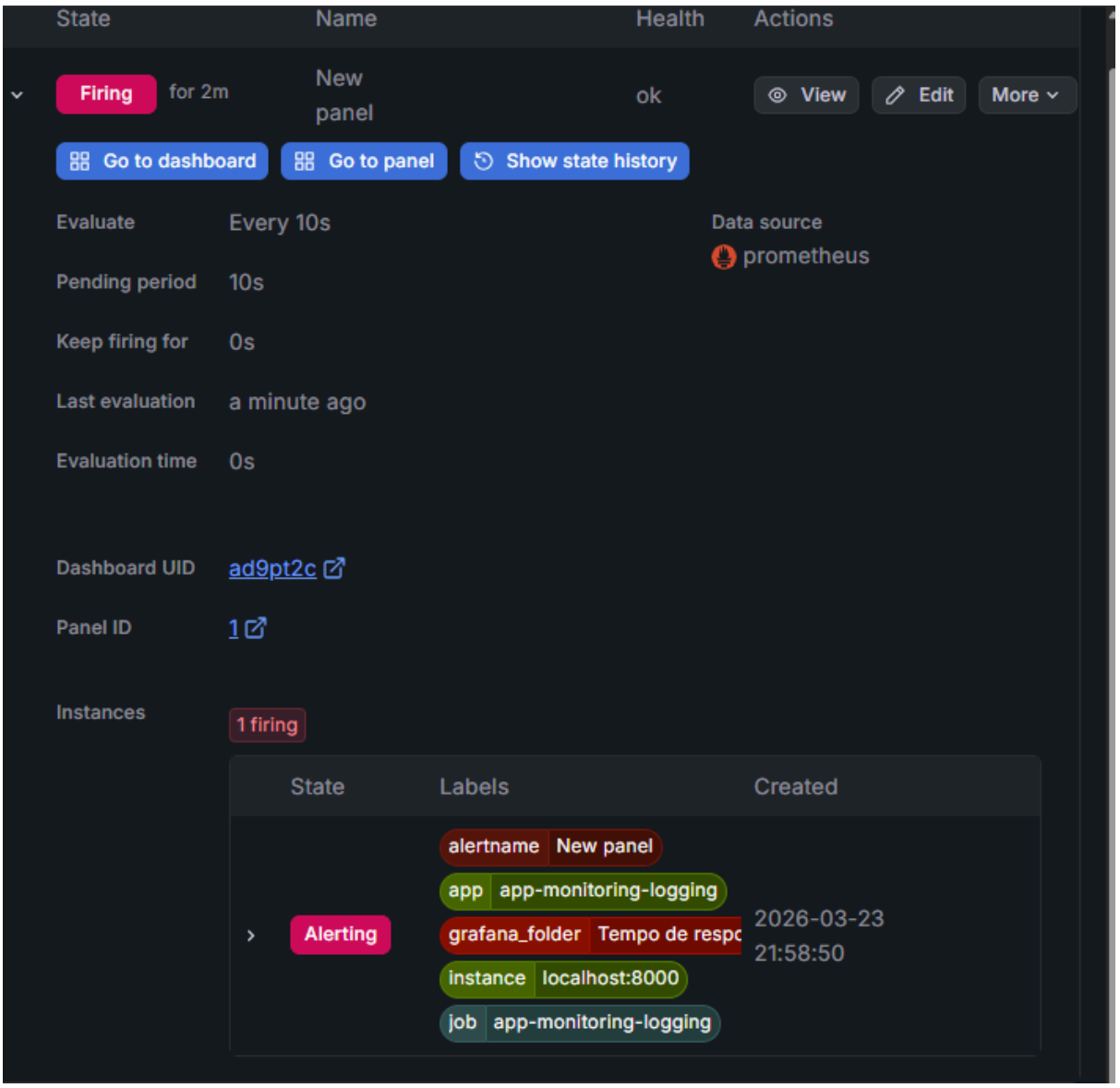


Simulação de lentidão

Exercício 5

New panel
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
21:55:30
21:56:00
21:56:30
21:57:00
21:57:30
21:58:00
21:58:30
21:59:00
21:59:30
22:00:00
.processing_seconds_sum", app="app-monitoring-logging", instance="localhost:8000", job="app-monitoring-logging"}


C
Threshold
Alert condition
Input
A
Is above 1
iame__="request_processing_seconds_sum", app="app-monitoring-logging", instance="localhost:8000", job="app-monitoring-logging"}
1
Firing



Firing pois foi testado o cenário com lentidão no sistema de 2s (acima de 1s)